

# Theoretical Foundations of Reinforcement Learning:

## Week 4

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### SARSA and Q-learning

Both SARSA and Q-learning are model-free TD methods for estimating action-value functions. They differ in exactly one place: how the next action is chosen in the update target.

#### Update Rules

**Q-learning:**

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[ R + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

**SARSA:**

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma Q(s', a') - Q(s, a)]$$

where  $a'$  is the action actually selected by the current policy in state  $s'$ .

The name SARSA comes from the tuple it uses:  $(s, a, r, s', a')$  — current state, action taken, reward received, next state, next action taken.

#### Derivation from Bellman Equations

Both updates are TD approximations of the Bellman equations for  $Q$ .

The Bellman expectation equation for  $Q_\pi$ :

$$Q_\pi(s, a) = \sum_{s'} P(s'|s, a) \left[ R(s, a, s') + \gamma \sum_{a'} \pi(a'|s') Q_\pi(s', a') \right]$$

SARSA approximates this by sampling: instead of summing over all  $s'$  and  $a'$ , it observes one transition  $(s, a, r, s', a')$  and treats  $r + \gamma Q(s', a')$  as the target. This is the on-policy TD target — the next action  $a'$  is drawn from  $\pi$ .

The Bellman optimality equation for  $Q^*$ :

$$Q^*(s, a) = \sum_{s'} P(s'|s, a) \left[ R(s, a, s') + \gamma \max_{a'} Q^*(s', a') \right]$$

Q-learning approximates this by sampling: it uses  $r + \gamma \max_{a'} Q(s', a')$  as the target regardless of what action the behaviour policy actually takes. This is the off-policy TD target — always assuming the best action will be taken next.

## On-policy vs Off-policy

Q-learning is **off-policy**. The behaviour policy (e.g.  $\varepsilon$ -greedy) collects experience, but the update always uses  $\max_{a'}$ , which corresponds to a fully greedy target policy. Q-learning directly estimates  $Q^*$  regardless of how data is collected.

SARSA is **on-policy**. The same policy that collects experience also determines  $a'$  in the update. SARSA estimates  $Q_\pi$  for the policy currently being followed, including its exploratory actions.

## Practical Consequence

Consider a cliff-walking task where falling off a cliff gives a large negative reward. An  $\varepsilon$ -greedy agent occasionally takes random actions near the cliff edge.

Q-learning learns the optimal path (close to the cliff) because its update assumes greedy behaviour. However, during training it occasionally falls due to  $\varepsilon$ -greedy exploration, incurring large penalties.

SARSA accounts for the fact that the agent will explore. It learns a safer path away from the cliff to avoid the risk of random actions causing falls. SARSA converges to the optimal policy for the  $\varepsilon$ -greedy agent, not the purely greedy one.

In environments where exploration-time safety matters, SARSA is preferred. When the goal is to learn the globally optimal policy regardless of training behaviour, Q-learning is preferred.

## Summary

|                       | SARSA                    | Q-learning                       |
|-----------------------|--------------------------|----------------------------------|
| Update target         | $R + \gamma Q(s', a')$   | $R + \gamma \max_{a'} Q(s', a')$ |
| Policy type           | On-policy                | Off-policy                       |
| Converges to          | $Q_\pi$ (current policy) | $Q^*$ (optimal policy)           |
| Safer during training | Yes                      | No                               |